

 **Write Model (Command)**

Action

Submit SQL 

Student

SV001 — Nguyễn Văn A 

Details

SELECT * FROM students

↑
EVENT BUS
↓

 **Read Model (Query)**

Total Submissions	0
Correct Answers	0
Active Contests	0
Top Scorer	—
Last Event	—
Projection Sync	Up to date

 Event Store

No events yet. Execute a command above.

 **Trong KBLab LMS:** Hiện tại hệ thống dùng shared database (anti-pattern). Với CQRS: write model xử lý submissions/scoring (consistency cao), read model phục vụ leaderboard + statistics (eventual consistency, cache-friendly). Event Sourcing cho phép rebuild leaderboard từ event history — rất hữu ích khi cần kiểm tra gian lận cuộc thi.